Module 15: Population and Systems Ecology - Learning Questions

1. What changes population size?
2. How do exponential and logistic growth differ?
3. What does carrying capacity mean?
4. How do density-dependent and density-independent factors differ?
5. Why do food chains have energy limits?
6. What does the 10 percent rule describe?
7. How do producers, consumers, and decomposers interact?
8. Why does energy flow but matter cycle?
9. How can feedback regulate a population?
10. What evidence would show a population is near carrying capacity?
11. Assume 10 percent energy transfer per trophic step. Starting with 20000 kJ in producer biomass, calculate the primary-, secondary-, and tertiary-consumer energy budgets. State why the assumption is approximate.
12. A herd grows from 40 to 90 in two years, then from 198 to 205 in two years. Compare the annual numerical increases and the increases relative to each starting count. What additional data would test a resource-limit explanation?